

Evidence Without Conclusion: Evaluating Evidence Acquisition and Inference in LLM Agent Societies

Anonymous authors

Paper under double-blind review

Abstract

Autonomous LLM agents are increasingly proposed for work that turns on noticing when a hidden ground truth has changed: monitoring systems, analysing experiments, running scientific workflows. An agent can fail at this in two quite different ways — it can fail to collect the evidence the conclusion needs, or it can fail to draw the conclusion from evidence it has already collected — and an evaluation that scores only the final answer cannot tell the two apart. We separate them. Observer Zero is an instrumented world with fictional physics in which LLM scientist agents choose measurements, exchange letters and maintain explicit probabilistic beliefs while a physical constant changes covertly mid-run, with the true state known only to the simulator. Measurement noise is seeded by (world, instrument, trial), so societies can be re-run against identical universes, and a non-LLM sequential change-point detector — given only the measurements the agents themselves chose to take, and no world constants, onset day or shift magnitude — estimates what was knowable from each run. Across two studies, the second pre-registered and frozen before any confirmatory data were seen, this decomposition localises the failure at one link. The detector finds the shift in 42 of 42 Study 2 runs, and the agents’ own measurement choices supported a stronger signal than a fixed reference schedule covering every instrument in the world; yet one agent of 276 concluded a physical law had changed, and that rate does not move anywhere along a monotone gradient in available evidence. Two further results concern the population. Communication here is voluntary and measured rather than scheduled: across 7,680 agent-days of opportunity, homogeneous grounded societies produced zero voluntary communications. When one agent from a different model family is introduced as a catalyst, nothing propagates beyond the agents it addresses directly, and of the twenty deliveries of unsupported claims originating with that agent, eighteen were incorporated into grounded agents’ beliefs and none were challenged. Evaluations that score conclusions without measuring what evidence was available, or that read communication volume as collective competence, will miss all three.

1 Introduction

LLM agents are increasingly proposed and built as autonomous investigators — debugging systems, analysing experiments, monitoring infrastructure, running scientific workflows. These settings share a structure. The agent observes evidence generated by a hidden ground truth, and must decide, among mundane and radical explanations, what changed. This is a detection problem the agent poses to itself: nothing labels the anomaly, and the space of candidate causes is the agent’s own.

Evaluating agents on it is harder than it looks, for a reason that is methodological rather than empirical. An agent that fails here has failed in one of two quite different ways: it did not collect the evidence the conclusion needs, or it did not draw the conclusion from evidence it had already collected. **Existing evaluations rarely distinguish the two**, because a benchmark that scores only the final answer cannot. The distinction matters

because the remedies diverge — the first is a data-collection problem, fixed by better tool use or a longer horizon; the second is an inference problem and is not.

We separate them by pairing a closed artificial world with a reference detector that is not a language model. A closed world gives the experimenter *perfect* ground truth while the agents hold only observations, instruments, memories and testimony: every belief can be scored against reality, every evidence citation traced or exposed as fabricated, and — the property this paper turns on — the world’s laws can be changed covertly, mid-run, so that whether anyone notices becomes a measurement rather than an interpretation. The detector supplies the missing quantity: a sequential change-point test run on exactly the measurements the society chose to take, with no privileged access to the world’s constants, the onset day, the shift magnitude, or which instrument class carries the signal. The difference between what it extracts from the agents’ own notebook and what the agents concluded from the same notebook estimates interpretation quality, isolated from data collection.

The same environment lets us treat a second architectural assumption as a measurement. Multi-agent LLM systems are usually built with communication scheduled, routed or required, which makes the question of whether agents would communicate at all structurally invisible. Here communication is voluntary throughout: no live arm contains scripted communication and no scheduler requires an agent to speak, so initiation, silence and what a channel carries are dependent variables.

The evidence comes from two studies in one such world, Meridian. Study 1 (Lamb, 2026) places two-agent societies in worlds where a physical constant silently shifts, varying model tier, provider, and the prompt’s instruction to prefer mundane explanations; it establishes the phenomenon and validates the instrument. Study 2, reported here for the first time, freezes a pre-registered design and scales to eight-agent societies, varying society size, the availability of a public institution, and the presence of a single agent drawn from a different model family. Study 2 depends on four Study 1 findings, chiefly that the grounded model is grounded and that its spontaneous communication baseline is genuinely zero, so the two are analysed together, with Study 1 results reported only where they carry the argument.

1.1 Contributions

An instrumented environment and a frozen evaluation platform: a deterministic world with fictional physics, instrument classes designed for causal discrimination rather than mere detection, power-analysed covert interventions, information-flow security enforced by type so that no prompt-constructing code can receive ground truth, seeded batteries, versioned manifests, a thirteen-class hypothesis taxonomy and a provenance-checked claim evaluator.

A three-level decomposition that estimates what was knowable. The non-LLM change-point detector receives no privileged world information and carries a built-in negative control, one instrument class being insensitive to the manipulated constant by construction. Comparing a fixed reference schedule over every instrument (L1), the society’s as-produced measurements (L2), a budget-matched subsample (L2d) and the society’s conclusions (L3) separates measurement-policy quality from data quantity from interpretation quality. We offer this as a reusable evaluation pattern, not only as an analysis of these runs.

Interpretation failure measured against an evidence gradient rather than against a single sufficiency claim, which would invite the reply that the threshold may lie above the level tested. A monotone gradient in available signal against a flat conclusion response locates the failure in what was concluded rather than in what was available or gathered.

Voluntary communication as a measured quantity, and what a catalysed channel carries: a zero-initiation baseline observed three independent times, then claim propagation with an origin split and provenance checking, in which the channel reliably carries unsupported claims and cascade depth never exceeds one.

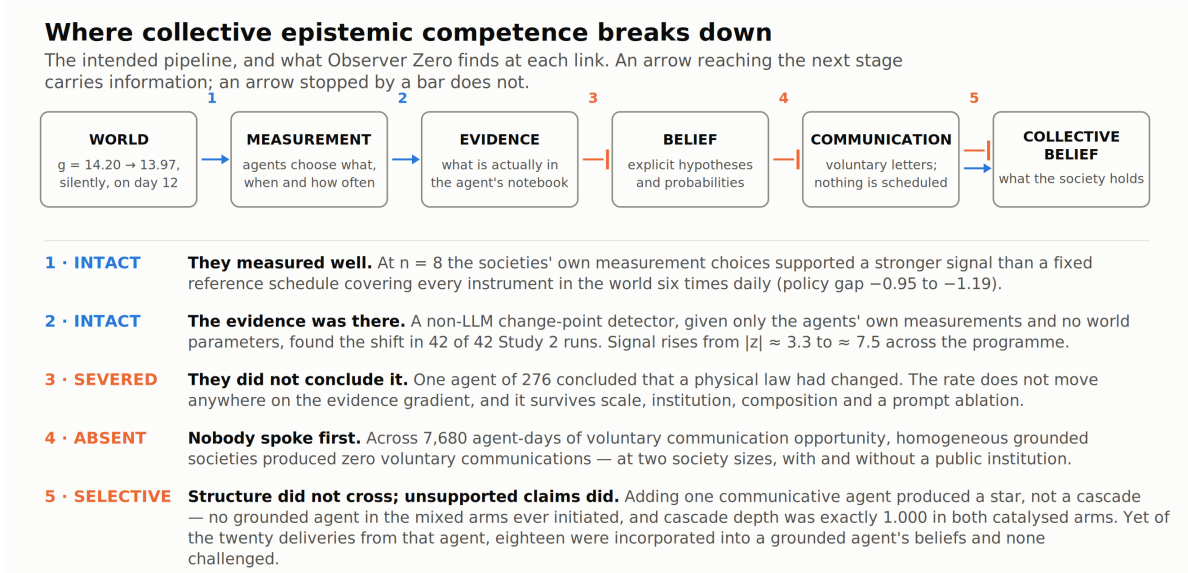


Figure 1: Where collective epistemic competence breaks down: the intended pipeline, and what Observer Zero finds at each link. The pipeline runs world $\rightarrow$ measurement $\rightarrow$ evidence $\rightarrow$ belief $\rightarrow$ communication $\rightarrow$ collective belief, and is intact from world to evidence, severed from evidence to belief, largely absent from belief to spontaneous communication, terminating at depth 1 from communication to network, and — the one place it works efficiently — open from communication to peer belief, for claims with no evidence behind them.

1.2 What we do not claim

That LLM agents under-use evidence is established elsewhere and at larger scale (Ríos-García et al., 2026); the contribution here is not the phenomenon but its location. We do not claim to have identified the ceiling’s cause, only to have made the two easiest explanations untenable; nor a general property of mixed agent populations, the activation results resting on one composition ratio and one model pairing; nor that errors amplify along agent chains, since nothing here propagated far enough to amplify and the opposite result has been reported (Jamshidi et al., 2026).

2 Related work

Evidence use, and instrumented worlds. That LLM scientific agents fail to use the evidence they generate is documented at scale: Ríos-García et al. (2026) evaluate agents across eight domains in more than 25,000 runs and report evidence non-uptake in 68% of traces, with only 26% exhibiting refutation-driven belief revision (Bisht et al., 2026). We take this as our starting point rather than our finding; neither work changes the environment’s rules mid-run, computes the statistical strength of the evidence collected, or separates a failure to detect from a failure to interpret. Several benchmarks do place agents in worlds whose structure must be discovered by experiment (Gandhi et al., 2025; Jansen et al., 2024; Wiemann et al., 2026; Zheng et al., 2026), but all are single-agent and in all of them the world’s law is non-canonical *from the start* — a task-generation device rather than a covert change during the run. The closest methodological precedent is STOCKTAKE (Deb & Krishnan, 2026), which drives what its authors call a fair oracle on the identical observation stream the agent receives. The move is the same as ours; the joint is different. **STOCKTAKE isolates detection $\rightarrow$ action; we isolate detection $\rightarrow$ belief** — ours notice and then conclude wrongly, which is harder to attribute because no action is available to reveal it. One difference runs in our favour: their oracle uses the environment’s true generative parameters, whereas our detector receives none (Engländer et al., 2026). On the individual side, belief revision under disconfirming evidence has been probed mainly through propositional tasks (Jhaveri et al., 2026) and the in-context locking Luo et al. (2025)

identify in ultra-long-horizon tasks — all on verbal evidence, none on noisy sensor data the agent chose to collect. One thing that literature does *not* support is worth stating, because our results are liable to be read as supporting it: larger models are *more* coherent with Bayes’ theorem, not less (Imran et al., 2025).

The voluntary-communication gap. Generative agent societies have grown from Park et al.’s (2023) sandbox to million-agent simulators (Yang et al., 2024), but their epistemic object is almost always opinion or convention — quantities with either no ground truth or a fixed one (Ashery et al., 2025). The field has developed a substantial self-critique, which we treat as our framing rather than as an incoming objection: audits against six validity principles find 90.7% of studies violate at least one (Zhou et al., 2025), and generative agent-based modelling has inherited little of agent-based modelling’s validation discipline (Larooij & Törnberg, 2025). Crucially, the field’s own communication survey (Yan et al., 2026) describes **no system in which agents autonomously decide whether to communicate at all**, and where the affordance exists no take-up or silence rate is reported. This is why our zero baseline is unmeasured territory rather than merely unreported: in a framework that schedules or requires communication, the quantity is structurally invisible.

Contamination and cascades. That LLM agents adopt peers’ claims is well quantified — harmful revision rising from 15.6% to 62.9% under unanimous wrong peers against beneficial revision from 32.7% to 51.5% under correct ones (Qu et al., 2026), with flip rates of 0.48 to 0.63 elsewhere (Cho et al., 2025) — but these are dyadic measurements with no network, no persistent belief state and no provenance. Error propagation along three-agent chains shows *attenuation* rather than amplification (Jamshidi et al., 2026), which makes “errors amplify through agent chains” an unsafe premise. The nearest neighbour to our contamination result is CoSim (Lin et al., 2026), whose misinformation is injected exogenously rather than generated inside the society. Two cautions against our own reading: some measured conformity may be a prompt artefact requiring no social speaker at all (Hu & Qu, 2026), and false claims presented as *presupposed* rather than asserted are accommodated far more readily (Cheng et al., 2026). The classical network-epistemology results supply the null expectations — informational cascades (Bikhchandani et al., 1992) and the finding that *less* connected communities can converge on truth more reliably (Bala & Goyal, 1998; Zollman, 2007), itself not robust across parameter values (Rosenstock et al., 2017) — but all of them presume that what flows through the channel is evidence. Appendix E gives this section in full, with the remaining literature.

3 The Meridian environment and confirmatory design

3.1 The world, the intervention and the agents

Meridian is a small settlement simulated in thirty discrete days by a deterministic engine. Measurement noise is keyed by the triple (world seed, instrument, trial index), so trial k on instrument i returns the same value in every run sharing that seed regardless of how the agents behave: evidence is a fixed property of the world, and a society can be re-run against an identical universe — the property every paired-by-seed analysis here depends on. The physics is fictional but structurally familiar: pendulum period is $T = 2\pi\sqrt{L/g}$ with $g = 14.20$ in fictional units, and crystal resonator frequency is $f = s\sqrt{R}$, insensitive to gravity by construction. Fictional constants prevent agents shortcutting discovery with memorised terrestrial physics. Relative measurement noise is 1%.

Each inhabited site hosts one pendulum and one resonator, which is the causal discrimination structure: a gravity change moves pendulums at every site and no resonators; a site-local environmental cause plausibly moves co-located instruments of both kinds; a single-rig fault moves one. The observation stream therefore supports diagnosis, not merely detection. Study 1 ran two sites and four instruments; Study 2 added six sites for sixteen, with the four Study 1 instruments unchanged in identity and noise stream.

Three scenarios were power-analysed before either study. In `gravity_shift`, g moves $14.20 \rightarrow 13.97$ on day 12, roughly 0.82% on pendulum period, tuned to be detectable-but-not-trivial under a realistic measurement schedule; in `instrument_fault` (Study 1 only) the laboratory pendulum reads $\times 1.008$ from day 12 while the physics is unchanged; in control, nothing happens. **Nothing about anomalies, interventions, or the existence of a simulation appears in any prompt or schema**, and hypothesis content is entirely agent-generated.

Agents hold personas with qualitative epistemic profiles and no numeric priors over exotic hypotheses. Each sees only its own instruments’ results and the messages it sends or receives, and per day chooses exactly one bounded action: run 1–12 trials on an owned instrument, write to a colleague, post to the bulletin where one exists, review its beliefs, or rest. It maintains an explicit belief state — self-generated hypotheses with probabilities plus a residual summing to one — and every update cites the event ids it rests on, which makes the attribution rule deterministic rather than inferential. **Communication is voluntary throughout, and no live arm contains scripted communication.** This is what makes the initiation results measurable at all, and it was maintained even where relaxing it would have de-confounded activation more cheaply: injecting a single scripted request into a silent society was considered and explicitly declined. Prompt builders structurally accept only an **AgentView** type carrying whitelisted observations, so no prompt-constructing code can receive world rules or ground truth, and an audit scans every stored prompt for privileged tokens; all 150 Study 1 and 85 Study 2 runs audited clean.

3.2 Frozen conditions and arms

Prompts, personas, engine constants and model settings are hashed into a manifest stamped into every run artifact. Study 1 ran under policy v0.1, its pre-registered ablation arm under a manifest differing by exactly one line of the belief prompt; Study 2 ran under v0.2. Both used ten world seeds (1000–1009) paired across scenarios, 30 days, temperature 1.0, and no provider exposes a sampling seed, so sampling variance is treated as part of measured society variance rather than eliminated.

Table 1: **Study 2 arms**, per the canonical specification in amendment A2. Each seed runs once in `gravity_shift` and once in `control`.

Arm	Composition	Institution	Runs
A	2 × sonar-pro	letters	20
B	8 × sonar-pro	letters	20
C	8 × sonar-pro	bulletin	5 → 20 by rule
D	7 × sonar-pro + haiku in the Theo slot	bulletin	20
E	7 × sonar-pro + sonnet in the Theo slot	bulletin	20

The minority persona slot is fixed: D and E place the minority model in the same Theo slot with identical persona text, so D – E varies the model and nothing else, and a test enforces it.

3.3 Pre-registration and the discipline actually applied

Seeds 1000–1009 were quarantined in code until the freeze commit, and the design froze at a tagged commit (hash and tag withheld for anonymity). After the final adversarial pass only *design-failure* fixes were permitted — a hypothesis unevaluable as written, an endpoint not computable from what is logged, a contradiction between sections — and not a better threshold, a cleaner definition, an extra metric or an extra arm. **No activation endpoint, credence or rate was computed until every arm was complete**, with one forced exception: evaluating arm C’s extension trigger required counting bulletin posts, which also revealed C’s letters, and it is recorded as such rather than presented as a later discovery. The test the decision table had to pass was: *for any plausible result, post-results the author can look here and know exactly how to analyse it*. Appendix A gives the nine frozen hypotheses with their verdicts, ten of the decision table’s eighteen rows, the two design decisions the freeze rule cost the study, and the position against the six PIMMUR validity principles of Zhou et al. (2025), who find 90.7% of LLM-society studies violate at least one.

3.4 Evaluation

Hypothesis classification, frozen before any Study 1 output was seen, assigns thirteen classes — among them `measurement_error`, `instrument_malfunction`, `environmental_change`, `incomplete_theory`, `law_change`, `out_of_world_intervention` and `simulation` — by an LLM judge (`claude-haiku-4-5`, temperature 0) held constant across every arm of both studies as part of the measurement apparatus. Pre-registered scoring: *detection* is anomaly-class mass above 0.5 in a belief snapshot; *strict* gravity diagnosis is `law_change` or

out_of_world_intervention dominant in the final state; *lenient* also accepts world-level natural causes. A separate judged pass asks whether each agent committed to a specific onset day and if so which, using the same frozen judge and prompt in both studies, and every factual evidence claim is classified against a closed source ontology with a deterministic lexicon tripwire for nonexistent sources as a model-free cross-check.

3.5 Endpoints

Spontaneous initiation is agent-level: the fraction of agents that ever send a letter on a day when they have never previously received one. Second-order activation requires an agent that had previously received a letter to open a new edge to one that had never written to it, with agents that ever initiated spontaneously excluded as seeds. Claim attribution is deterministic and citation-primary — a belief hypothesis whose evidence array includes a delivery event id is attributed to that claim, and **timing alone never attributes** — with claims split by origin into `FIRST_PARTY` and `RELAYED_FROM_ANOTHER`. Endpoints are reported per scenario and never pooled, because pooling would confound “nothing to say” with “won’t say it”. Appendix B gives every definition in full.

3.6 The three-level detector benchmark

A non-LLM sequential change-point detector estimates what was knowable. It establishes a baseline from the first ten days, then flags the first day on which the cumulative post-baseline mean exceeds 2.5 standard errors. *L1 potential* is every instrument in the world on a fixed reference schedule of six trials daily, and requires the simulator since it measures counterfactual instruments; *L2 as-produced* is exactly the measurements the society chose to take; *L2d* is L2 subsampled to an $n = 2$ -equivalent budget over 200 seeded draws, pricing raw data quantity alone; and *L3* is what the society decided. $L1 \rightarrow L2$ is measurement-policy quality, $L2$ vs $L2d$ is data quantity, and $L2 \rightarrow L3$ is **interpretation quality**.

Two properties matter. **The detector receives no privileged world information** — not the shift magnitude, not the onset day, not the world constants, not which instrument class carries the signal — consuming only per-instrument series of (day, observed value) pairs. And **it carries a built-in negative control**: resonator frequency is insensitive to gravity by construction, so every resonator flag in a gravity world is a false alarm, which bounds the detector’s own error rate by construction; the control-world flag rate is reported under limitations below. Any claim that a society “detected” something must clear that floor, not merely zero.

4 Results

Reported as questions rather than as H1–H7, so that each result closes off an alternative explanation for the one before it; Appendix A maps them back to the frozen hypotheses. Study 1 results appear where they carry the chain, compressed to the four findings Study 2 depends on; the remainder are in the Study 1 report (Lamb, 2026).

4.1 Data quality

All 85 Study 2 runs completed with no re-runs, the stale-final-state rate was zero in every arm against a pre-registered 10% flag, and the leak audit was clean, so the pre-registered primary and sensitivity analyses are identical by construction and reported as one.

4.2 How much evidence did the agents actually gather?

The first objection to everything that follows is that the shift may simply have been too small to find. If so, declining to conclude *law change* would be correct Bayesian behaviour and the ceiling would be a statement about the scenario rather than about the agents. Three independent arguments close this off, and only two use the detector.

The task is solvable without any detector at all. Study 1’s scripted mock society — arm B0, which always replicates, always blindly, and applies textbook sequential statistics — diagnoses gravity strictly in 10 of 10 gravity worlds on the any-agent criterion and stays quiet in 7 of 10 control worlds. The observation stream therefore contains sufficient information for a specified scientific procedure to discriminate the conditions — **without establishing that an autonomous LLM agent should reach 10 of 10** — using no instrument a sceptical reader would have to accept.

The signal in the agents’ own notebooks rose, and the conclusion did not follow.

Table 2: **L2 as-produced, gravity_shift worlds, ten-day baseline.** Mean maximum pendulum $|z|$ over runs, and runs in which the detector flagged the change.

Condition	n	mean $ z $	flagged	law-change conclusions
Study 1 B1 haiku	2	3.30	8/10	0
Study 1 B2 sonnet	2	3.88	7/10	0
Study 1 B3b sonnet-nmp	2	4.49	9/10	0 (one in a <i>control</i> world)
Study 1 B3a sonar	2	5.00	10/10	0
Study 2 arm A sonar	2	5.77	10/10	0 of 20 agents
Study 2 arm B sonar	8	7.28	10/10	1 of 80 agents
Study 2 arm C sonar	8	7.09	2/2	0 of 16 agents
Study 2 arm D	8	7.42	10/10	0 of 80 agents
Study 2 arm E	8	7.52	10/10	0 of 80 agents

Signal strength rises by a factor of 2.3 across the programme and detection rises from 7 of 10 runs to 10 of 10, while **the law-change conclusion rate does not move off the floor at any point on that gradient.** This is stronger than sufficiency: a single sufficiency claim invites the reply that the threshold might lie above $z \approx 7$, whereas a monotone gradient with a flat response shows no point on the observed range at which the behaviour changes.

And their measurement policy was not the problem. The shift was detectable in 42 of 42 Study 2 gravity_shift runs, typically from day 12 of 30, and remains so in 99–100% of draws when downsampled to a two-agent budget, so this is not a benefit of scale. At $n = 8$ the L1 $\rightarrow$ L2 gap is *negative* in every arm.

Table 3: **the three-level detector benchmark.** A negative policy gap means the society’s own measurement choices supported a stronger signal than the fixed reference schedule.

Arm	L1 reference	L2 as-produced	policy gap	L2d at $n=2$ budget
A ($n=2$)	6.33	5.77	+0.56	100% of draws
B ($n=8$)	6.33	7.28	−0.95	99%
D ($n=8$)	6.33	7.42	−1.09	99%
E ($n=8$)	6.33	7.52	−1.19	100%

The decomposition isolates the failure precisely: L1 $\rightarrow$ L2, measurement-policy quality, better than reference; L2 $\rightarrow$ L3, interpretation quality, total. The diagnostic is not constructed to convict the agents — run on Study 1’s per-agent streams it estimates onset at median day 10.5 on haiku’s erratic schedules against day 12.0 on sonnet’s more systematic ones, exonerating one arm and convicting another. Its robustness to the baseline window and its false-alarm floor are reported under limitations.

4.3 Did they interpret it?

Almost never, in either study. In Study 1 all live arms detected intervention anomalies transiently in 90–100% of gravity worlds, yet the strict gravity diagnosis rate was 0 of 10 runs in every arm, 0 of 40 overall across 80 agent-final belief states — a rule-of-three 95% upper bound on runs of 7.5% — while instrument-fault worlds were correctly diagnosed by at least one agent in up to 70% of runs. **Diagnostic capability exists, inside the mundane vocabulary; the boundary is specifically at world-level revision.** In Study 2 the strict rate is 0 at $n = 8$ in every arm.

Per arm the counts are 0 of 20 (A), 1 of 80 (B), 0 of 16 (C), 0 of 80 (D) and 0 of 80 (E), with mean credence on the class never above 0.0065 (Appendix D). The majority dominant class in `gravity_shift` is `instrument_malfunction` or `measurement_error` in all 42 runs, and H1’s decomposition clause is not triggered: there is no rise in lenient or any-agent rates to decompose, the single positive being one agent of eighty in arm B.

The ceiling is not one failure, and this is the second Study 1 result Study 2 depends on. Haiku exhibits a *generation* failure — gravity or constant-change hypotheses appear anywhere in only 2 of 20 final states — while sonnet exhibits a *commitment* failure, such hypotheses entering 15 of 20 trajectories, peaking as high as $p = 0.85$, and being abandoned before the final state in nearly all. The closest approach reached $p = 0.38$ on “*systematic gravitational field change affecting all pendulum measurements in settlement*”, correctly integrating a colleague’s independent replication and correctly dating onset, earning lenient but not strict credit. Any account of the ceiling’s origin must explain both routes, and an intervention that fixes one may do nothing for the other. No agent in either study produced a simulation-class hypothesis as a dominant belief; more strikingly, none adopted the substantially weaker hypothesis that a physical law of their world had changed.

4.4 Was the prompt prior responsible, or was the society too small?

Neither. The most obvious sceptical reading — *you told them to prefer mundane explanations* — was tested directly by an ablation removing that single line from the belief prompt (Study 1 arm B3b). It did not produce strict gravity diagnoses (0 of 10); it tripled lenient world-level commitment, 1 of 10 $\rightarrow$ 3 of 10; and it degraded control-world calibration, with one control agent reaching `law_change` dominant — **the only law-change verdict in Study 1, produced in a world where nothing had changed**. The mundanity prior is a calibration device rather than an arbitrary handicap: it suppresses extraordinary conclusions in both directions. Note the interaction with the evidence gradient: B3b also had *stronger* evidence than the unablated sonnet arm ($|z|$ 4.49 against 3.88). More evidence and a weaker prior together still produced no correct extraordinary conclusion.

Nor did scale help. Arm A ($n = 2$) and arms B–E ($n = 8$) are indistinguishable at the strict endpoint, despite the eight-agent arms carrying roughly four times the observation stream and a correspondingly stronger signal. The ceiling is not a property of society size, institution, or composition.

4.5 Do homogeneous grounded agents spontaneously communicate?

No, and this is the most heavily replicated result in the programme. It is also the third Study 1 result Study 2 depends on: arm B’s zero is only interpretable because Study 1 established that this model produces no letters in a two-agent society. Arm B’s rates are not merely near zero but exactly zero — no letter in 20 runs and 160 agent-runs — and the same holds in Study 1’s two-agent letter arm (30 runs) and in all three pilot conditions, at both society sizes and with and without a bulletin (Appendix D). Summing the homogeneous grounded conditions gives **7,680 agent-days of voluntary communication opportunity and zero voluntary communications**. Agent-days measure exposure, not independent trials, so the interval is attached to the agent-run, the unit the endpoint is defined on: zero initiations in 256 agent-runs, exact 95% CI [0, 1.43%]. This is not deliberation ending in refusal — of 180 action reasons recorded in the pilot’s two-agent bulletin condition, exactly one even names the bulletin or the colleague, and that one is a false-positive match.

Scope, precisely. Spontaneous communication was absent in every pre-registered letter condition, and across the 320 agent-runs of the two mixed-composition arms no grounded agent ever initiated. One exception is reported rather than smoothed: in arm C’s `gravity_shift-seed1001` a pure-sonar dyad forms, one spontaneous initiator in that arm’s 40 agent-runs, with all eight agents verified `sonar-pro` in the manifest before this was believed. One available reading is that a chatty peer *suppresses* grounded initiation rather than enabling it; untested here.

4.6 Does adding one communicative agent create a network?

It creates communication. It does not create a network. H3’s prediction is met in its strongest available form, B’s rate being exactly zero, but the mechanism is narrower than “a minority agent activates a network”: the +0.125 is one agent in eight, and the same agent every time — Theo, the minority, in all 20 runs of D and all 20 of E. What distinguishes D is that the seed’s letters get answered.

Cascade depth is exactly 1.000 in every scenario of both catalysed arms, and arm B produced no letters to have a depth at all; reply rate given addressed is 0.771/0.775 in D against 0.167/0.267 in E, with cascade reach 0.314/0.329 and 0.200/0.186 (Appendix D). Nothing propagated beyond the agents the seed addressed directly, so under the pre-registered interpretation rule the result is **a star around the seed, not a cascade** (Appendix D, Figure 3). Second-order activation exists but is thin, and 2 of D’s 20 runs meet the active-network bar. Depth 1 was not a post-hoc observation: the pilot correction, issued before the freeze and before any confirmatory spend, recorded that depth was 1 in every pilot run — *“the network densified — more edges, but never deepened into a chain.”* The confirmatory phase tested a stated prediction.

4.7 What else the catalysed arms show

Three secondary results are summarised here and set out with their tables in Appendix D. **Convergence got worse, not better:** belief dispersion fell *more slowly* in D and E than in the silent arm B, a direction consistent across both scenarios and both contrasts, though no individual contrast reaches significance under an exact sign test and the forty seed-level comparisons share one baseline, so we do not pool them. **The seed behaves identically; the network’s response does not:** both minority models initiate in every run and the recruitment gap is dosage rather than eloquence (4.93 letters per recipient in D against 1.81 in E), but fabrication genuinely diverges — in the same slot, with identical persona text, on the same seeds, the minority produced 19 unsupported claims as haiku and 1 as sonnet, with eight discordant pairs in one direction and none in the other, **McNemar exact $p = 0.0078$** . The per-letter rates of 0.086 against 0.020 are reported against rather than in support of the finding: with a single event in arm E the interval on that ratio includes 1, so “a factor of four” is not defensible. This dissociation replicates Study 1 — the fourth Study 1 result Study 2 depends on — where judged fabrication was 24 of 60 agents for haiku and 9 of 60 for sonnet against **0 of 60 for sonar**, an absence rather than a reduced rate. **The public institution was essentially never used:** across the study the bulletin was written to exactly once, and that post was a delivery check rather than a finding, so it was not rejected as an epistemic commons so much as never conceived of as one.

4.8 What did communication transmit?

Unsupported claims, which were accepted. The judged pass on arm D used the frozen evaluator at temperature 0: 325 judge calls, **0 judge failures and 0 unjudged exposures**.

Table 4: **judged claim propagation, arm D.**

unsupported claims, FIRST_PARTY origin	20 — 19 by Theo (haiku minority), 1 by Jamie (sonar)
delivered exposures	21 (0 re-exposures)
runs containing any exposure stances	9 of 20 INCORPORATED INTO BELIEF 19 · IGNORED 1 · CORRECTED 1 · CHALLENGED 0 · ENDORSED 0
attribution basis	citation 17 · judge 4 · none 0
transmission / contamination	0.959 / 0.959 across the 9 runs with claims
unattributed belief changes	745

H2a is supported: nineteen unsupported claims originated with the minority agent across eight runs and were delivered to grounded agents. **H2b is supported, and not marginally:** eighteen of the twenty minority-origin deliveries reached **INCORPORATED INTO BELIEF** across three distinct sonar recipients — nineteen of all twenty-one exposures once the one grounded-origin claim is included — and not one exposure was **CHALLENGED**.

The attribution rule cut against the finding, not for it: seventeen of 21 attributions are citation-based, the recipient listing the delivery event id in its own evidence array rather than an evaluator inferring influence from adjacency, while the no-attribution-by-proximity rule discarded 745 belief changes because nothing cited them. Contamination also runs both ways — the one non-minority origin is a sonar agent asserting unsupported environmental quirks to Theo, who incorporated it — which is why the FIRST_PARTY / RELAYED_FROM_ANOTHER split was built before the freeze: without it this claim would have scored as a grounded agent fabricating, inverting the distinction the study exists to draw.

4.9 Did early onset back-dating persist at $n = 8$?

Yes, in every arm. This hypothesis was evaluated after the other six — its dating pass was never run during the confirmatory phase, and was run subsequently with the frozen judge and prompt over the same artifacts, with the omission, resolution and residual exposure recorded in Appendix C. Pooled at $n = 8$, **123 of 141 dated onsets precede the true onset of day 12, 0.872, 95% CI [0.806, 0.923]**; an exact one-sided binomial against a null of 0.5 gives $p \approx 1 \times 10^{-20}$, the more conservative of the two available tests, since counting early against late alone would give $p \approx 1 \times 10^{-24}$. Median dating error is negative in every arm, and arm-level detail is in Appendix D. H4’s frozen wording is a directional prediction about persistence, not a requirement that each arm reach significance independently.

Two things the frozen hypothesis did not anticipate are reported as observations and not promoted to endpoints. Unconditional date commitment is not scale-invariant: the rate of dating an onset in quiet worlds falls from 0.838 in Study 1 to 0.341 at $n = 8$, though Study 1’s figure was driven by its Anthropic arms while its sonar arm sat at 11/20, and arm A here — same model, same size, same seeds — is 11/20 exactly. And agents at $n = 8$ date an onset far less often, 19 of 20 in arm A against 141 of 256, which shares the prompt-version confound recorded in the synthesis.

4.10 Cross-study synthesis

Four things are true across both studies that neither establishes alone. **The evidence gradient is monotone and the conclusion response is flat** — Table 2 spans nine conditions from $|z|$ 3.30 to 7.52 while the law-change rate stays pinned at the floor, a shape that exists only because the two studies share a world, a detector and a judge. **Zero spontaneous communication is observed three independent times**, at two society sizes and with and without a public institution. **The fabrication dissociation recurs in a different design**, measured as a rate across whole societies in Study 1 and as a controlled substitution in a single persona slot in Study 2, sharing no analysis path yet separating the same model families in the same direction. And **back-dating replicates while unconditional commitment does not**: Study 1’s early-dating fractions of 0.79 to 0.85 become 0.872 pooled at $n = 8$, exactly as H4 predicted, while the related behaviour Study 1 called invariant “across all tested model and prompt conditions” (Lamb, 2026, p.9) attenuates from 0.838 to 0.341 across the same scale change. Reporting both directions is the point: one comparison confirms a prior claim and the other narrows it.

One limitation applies to the synthesis rather than to either study. Study 1’s B3a and Study 2’s arm A are both two-agent sonar societies on the same world seeds — literally the same universes — yet their as-produced signal differs, 5.00 against 5.77. Since the worlds are identical that gap is attributable entirely to what the agents chose to measure, and the prompt templates changed between the studies, so **cross-study comparisons of as-produced signal confound measurement policy with prompt version** and the gradient must not be described as a clean manipulation.

5 Discussion

5.1 Implications for deployed multi-agent systems

Three, at the strength of the evidence. Collective competence cannot be inferred from communication volume: arm D produced the most letters, the most edges, the most cascade beliefs — 17, against 0 in A, B and C and 1 in E — alongside the lowest independent-source count. The interpretation step should be

expected to be the weak one, so a monitoring system built from such agents would collect adequate evidence of a regime change and report an instrument fault — and evaluations that score data collection, or that score conclusions without measuring what evidence was available, will both miss this. And a quiet channel is not a safe channel: eighteen of twenty unsupported claims were incorporated in a society that never formed a network at all.

5.2 Communication capacity does not produce collective epistemic competence

The programme’s arms were built to add, one at a time, the things a population is supposed to need in order to know more than an individual: more observers, a shared public record, and a peer who talks. None moved the conclusion the world was built to elicit. What the talking peer produced was contamination, while dispersion in those same arms fell *more slowly* than in the silent counterfactual on identical worlds. The society that talked was the same epistemic system with a channel attached, and what the channel carried was the one class of content with no evidence behind it. **Adding communicative capacity to a population of LLM agents is not sufficient for collective epistemic competence, and in these societies it was not even neutral** — though one composition ratio, one model pairing and one world are not a general result.

5.3 Where the ceiling does not come from

Two studies make the two easiest explanations untenable: not insufficient evidence, since the gradient is monotone and the negative policy gap rules out a data-collection artefact; and not the explicit prompt prior, since removing the guardrail did not convert conservatism into insight but into false alarms — a design finding in its own right for deployed systems, which will inherit some version of that prior either explicitly or through training. What remains is a shorter list, and we do not adjudicate between its items: epistemic conservatism shared across contemporary models from training; the agent architecture and its evidence presentation; the experimental framing; or interactions among these.

5.4 Model choice as epistemic-culture choice

Holding architecture, prompts, personas and worlds constant, the choice of foundation model set collaboration rate, fabrication propensity and anomaly appetite more strongly than it set diagnostic accuracy. “Communicativeness” is not one trait, and treating it as a single dial — as one does when selecting a model for a multi-agent deployment — will mispredict at least two of the three. The same analysis narrows Study 1’s claim that unconditional date commitment survives “across all tested model and prompt conditions” (Lamb, 2026, p.9): that rate was largely a model effect, replicating within condition while attenuating across society size, so it is a robust property of a model in a setting rather than an invariant of the architecture. A related point holds for the temporal result — the detector places the change point at approximately the right day while the agents, reading the same series, place it systematically early — so the pipeline is not only *evidence present* → *conclusion absent* but, where a temporal judgement is made at all, *evidence present* → *temporal interpretation distorted*.

5.5 A serious alternative reading of the contamination result

The obvious interpretation is social: grounded agents deferred to a peer. Hu and Qu (2026) make that harder to sustain without a control we do not have. Holding the asserted answer fixed and deleting the explicit speaker, they find the no-source condition alone induces harmful revision in 66.5% of initially correct items across six models and seven datasets, against 10.3% under a plain re-ask. Revision should therefore be credited to a social effect only as an increment *above* what bare asserted text already produces, and every claim in our societies arrives inside an attributed letter from a named colleague. What survives is substantial: arm B is a control their design does not have, since no assertion is present at all, so the D – B contrast is *assertion present versus assertion absent*. That claims were incorporated, that none were challenged, and that recipients cited the delivery event id are all unaffected; what is at risk is the further inference that this

reflects social deference specifically. The missing arm is now specified — deliver identical claim content with the sender attribution stripped, and measure incorporation against the attributed condition.

5.6 Two results that must be read together

That every arm classifies as an independent ensemble and that unsupported claims contaminated three grounded agents look, at first reading, contradictory. They are not, and the tension is itself the finding: a society can fail every reasonable threshold for being an epistemic community — no spontaneous initiation, cascade depth 1, a third of agents producing any letter at all — and still transmit unsupported claims efficiently. **Low connectivity is not protective.** The classical result that sparse communication can *help* an epistemic community (Bala & Goyal, 1998; Zollman, 2007) — itself parameter-dependent (Rosenstock et al., 2017) — presumes that what flows through the sparse channel is evidence. Here what flowed was assertion.

6 Limitations

All results sit within one agent loop, one memory design and one prompt family, so invariance claims are invariant across what we varied and no further. Ten seeds per condition per arm give wide intervals on binary outcomes; the headline 0-of-40 is robust in direction but its exact 95% upper bound on runs is 8.8%, not zero. The institution null may be persona-scoped, the bulletin being read overwhelmingly by the journalist. Agents know Kuhn and Bostrom from pretraining, so Observer Zero does not test whether naive minds invent the simulation hypothesis. And no provider exposes a sampling seed, so run-level variance includes sampling variance, which is also part of the phenomenon.

On measurement: all judged metrics use an Anthropic judge over Anthropic and Perplexity agents, triangulated with a lexicon tripwire and a manual claim audit, but inter-judge agreement with a non-Anthropic judge remains future work, and exposure is *delivered*, not attended. The detector’s ten-day baseline is an analyst choice, clean because the onset is day 12; recomputed at 6, 8, 10, 12 and 14 days — the last two deliberately contaminated by post-onset observations — every gravity_shift run in every Study 2 arm is flagged at every setting, and the contaminated windows *degrade* the statistic, confirming ten days is the largest clean window, not a tuned one. In control worlds the detector flags 1–4 runs of 10 at mean $|z|$ 1.5–1.8, so it is not a perfect instrument, and it remains an instrument we specified rather than a ground truth. The detection criterion also counts self-error and environmental attributions as anomaly mass, which inflates “detection” for chatty models, though all arms are scored identically. Design confounds are set out in Appendix C, with the audit trail.

Broader Impact Statement

The results concern the reliability of LLM agents proposed for use as autonomous investigators, and two bear directly on deployment risk. A monitoring system built from agents of this kind would collect adequate evidence of a regime change and report a local instrument fault, so evaluations that score data collection, or that score conclusions without measuring what evidence was available, will certify such a system as working. And a channel that carries almost no traffic is not thereby safe. We report a failure to draw a warranted conclusion rather than a capability that could be misused, and the environment contains no real-world knowledge, so we see no dual-use concern. The chief risk of misreading this work is over-generalisation: the results are bounded by one world and the model versions in the manifests.

Reproducibility Statement

Measurement noise is keyed by (world seed, instrument, trial index), so any society can be re-run against an identical universe; prompts, personas, engine constants and model settings are hashed into a manifest stamped into every run artifact; and the hypothesis judge is one frozen model at temperature 0, held constant across both studies. The model code, the raw run artifacts and derived artifacts sufficient to reproduce every number here are deposited publicly; the deposit is withheld from this version for anonymity.

References

- Anderson, L. R. & Holt, C. A. (1997). Information cascades in the laboratory. *American Economic Review*, 87(5), 847–862.
- Ashery, A. F., Aiello, L. M. & Baronchelli, A. (2025). Emergent social conventions and collective bias in LLM populations. *Science Advances*, 11(20), eadu9368.
- Bala, V. & Goyal, S. (1998). Learning from neighbours. *Review of Economic Studies*, 65(3), 595–621.
- Banerjee, A. V. (1992). A simple model of herd behavior. *Quarterly Journal of Economics*, 107(3), 797–817.
- Bikhchandani, S., Hirshleifer, D. & Welch, I. (1992). A theory of fads, fashion, custom, and cultural change as informational cascades. *Journal of Political Economy*, 100(5), 992–1026.
- Bisht, H., Kumar, V., Jablonka, K. M., Mausam & Krishnan, N. M. A. (2026). Agentic AI scientists are not built for autonomous scientific discovery. arXiv preprint arXiv:2605.08956.
- Centola, D. & Macy, M. (2007). Complex contagions and the weakness of long ties. *American Journal of Sociology*, 113(3), 702–734.
- Cheng, M., Hawkins, R. D. & Jurafsky, D. (2026). Accommodation and epistemic vigilance: A pragmatic account of why LLMs fail to challenge harmful beliefs. arXiv preprint arXiv:2601.04435.
- Cho, Y.-M., Guntuku, S. C. & Ungar, L. (2025). Herd behavior: Investigating peer influence in LLM-based multi-agent systems. arXiv preprint arXiv:2505.21588.
- Deb, S. & Krishnan, A. (2026). STOCKTAKE: Measuring the gap between perception and action in LLM agents with a fair oracle. arXiv preprint arXiv:2607.13618.
- Engländer, L., Althammer, S., Üstün, A., Gallé, M. & Sherborne, T. (2026). Agents explore but agents ignore: LLMs lack environmental curiosity. arXiv preprint arXiv:2604.17609.
- Gandhi, K., Li, M. Y., Goodyear, L., Bhatia, A., Li, L., Bhaskar, A., Zaman, M. & Goodman, N. D. (2025). BoxingGym: Benchmarking progress in automated experimental design and model discovery. arXiv preprint arXiv:2501.01540.
- Hu, Y. & Qu, J. (2026). Most LLM conformity needs no speaker: Measuring the speaker-free floor in peer-pressure benchmarks. arXiv preprint arXiv:2607.05545.
- Imran, S., Kendiukhov, I., Broerman, M., Thomas, A., Campanella, R., Lamb, R. & Atkinson, P. M. (2025). Are LLM belief updates consistent with Bayes’ theorem? *ICML 2025 Workshop on Assessing World Models*. arXiv preprint arXiv:2507.17951.
- Jamshidi, S., Moradi Dakhel, A., Nafi, K. W. & Khomh, F. (2026). Hallucination cascade: Analyzing error propagation in multi-agent LLM systems. arXiv preprint arXiv:2606.07937.
- Jansen, P., Côté, M.-A., Khot, T., Bransom, E., Dalvi Mishra, B., Majumder, B. P., Tafjord, O. & Clark, P. (2024). DiscoveryWorld: A virtual environment for developing and evaluating automated scientific discovery agents. *Advances in Neural Information Processing Systems 37*, Datasets and Benchmarks Track.
- Jhaveri, A. R., GX-Chen, A., Sucholutsky, I. & Choi, E. (2026). Failing to falsify: Evaluating and mitigating confirmation bias in language models. arXiv preprint arXiv:2604.02485.
- Lamb, N. (2026). Observer Zero: Autonomous LLM scientists detect changes to their world but fail to conclude that it changed. Zenodo. <https://doi.org/10.5281/zenodo.21872780>
- Larooij, M. & Törnberg, P. (2025). Do large language models solve the problems of agent-based modeling? A critical review of generative social simulations. arXiv preprint arXiv:2504.03274.
- Li, Y., Naito, A. & Shirado, H. (2025). Systematic failures in collective reasoning under distributed information in multi-agent LLMs. arXiv preprint arXiv:2505.11556.
- Li, Y. & Tao, D. (2026). AI agents alone are not (yet) sufficient for social simulation. arXiv preprint arXiv:2603.00113.
- Lin, C., Jin, Y., Hu, K., Fan, W., Xiao, H., Wang, Y., Ying, Z. & Zhao, Z. (2026). You can’t fool us: Understanding the resilience of LLM-driven agent communities to misinformation. arXiv preprint arXiv:2605.17353.
- Liu, K., Xiong, G., Zhang, W. & Tang, S. (2026). Social networks of LLM agents. arXiv preprint arXiv:2607.03695.

- Lu, Y., Yang, S., Qian, C., Chen, G., Luo, Q., Wu, Y., Wang, H., Cong, X., Zhang, Z., Lin, Y., Liu, W., Wang, Y., Liu, Z., Liu, F. & Sun, M. (2024). Proactive agent: Shifting LLM agents from reactive responses to active assistance. arXiv preprint arXiv:2410.12361.
- Luo, H., Zhang, H., Zhang, X., Wang, H., Qin, Z., Lu, W., Ma, G., He, H., Xie, Y., Zhou, Q., Hu, Z., Mi, H., Wang, Y., Tan, N., Chen, H., Fung, Y. R., Yuan, C. & Shen, L. (2025). UltraHorizon: Benchmarking agent capabilities in ultra long-horizon scenarios. arXiv preprint arXiv:2509.21766.
- Niu, R., Shu, X. & Zhao, Y. (2026). Reliability-contagion feasibility in LLM multi-agent networks. arXiv preprint arXiv:2607.21912.
- O’Connor, C. & Weatherall, J. O. (2019). *The Misinformation Age: How False Beliefs Spread*. New Haven, CT: Yale University Press.
- Park, J. S., O’Brien, J. C., Cai, C. J., Morris, M. R., Liang, P. & Bernstein, M. S. (2023). Generative agents: Interactive simulacra of human behavior. *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST ’23)*.
- Qu, J., Fu, L. & Hu, Y. (2026). Easier to mislead than to correct: Harmful and beneficial revision in LLM conformity. arXiv preprint arXiv:2606.01637.
- Ríos-García, M., Alampara, N., Gupta, C., Mandal, I., Mannan, S., Aghajani, A. A., Krishnan, N. M. A. & Jablonka, K. M. (2026). AI scientists produce results without reasoning scientifically. arXiv preprint arXiv:2604.18805.
- Rosenstock, S., Bruner, J. & O’Connor, C. (2017). In epistemic networks, is less really more? *Philosophy of Science*, 84(2), 234–252.
- Vezhnevets, A. S., Agapiou, J. P., Aharon, A., Ziv, R., Matyas, J., Duéñez-Guzmán, E. A., Cunningham, W. A., Osindero, S., Karmon, D. & Leibo, J. Z. (2023). Generative agent-based modeling with actions grounded in physical, social, or digital space using Concordia. arXiv preprint arXiv:2312.03664.
- Wiemann, M. L., Smith, L. M., Melchior, P., Mishra-Sharma, S., Wilson, A. G., Izmailov, P. & Cuesta-Lázaro, C. (2026). DiscoverPhysics: Benchmarking LLMs for out-of-the-box scientific thinking. arXiv preprint arXiv:2605.26087.
- Wu, B., Zou, G., Wang, B., Zhao, H. & Shi, C. (2026). Ask now, use later: Benchmarking the proactivity gap in long-lived LLM agents. arXiv preprint arXiv:2605.28108.
- Yan, B., Zhou, Z., Zhang, L., Zhang, L., Zhou, Z., Miao, D., Li, Z., Li, C. & Zhang, X. (2026). Beyond self-talk: A communication-centric survey of LLM-based multi-agent systems. *Frontiers of Computer Science*.
- Yang, K., Peng, T.-Q., Lee, S. & Liu, H. (2026). Think-before-speak: From internal evaluation to public expression in multi-agent social simulation. arXiv preprint arXiv:2606.03137.
- Yang, Z., Zhang, Z., Zheng, Z., Jiang, Y., Gan, Z., Wang, Z., Ling, Z., Chen, J., Ma, M., Dong, B., Gupta, P., Hu, S., Yin, Z., Li, G., Jia, X., Wang, L., Ghanem, B., Lu, H., Lu, C., ... Shao, J. (2024). OASIS: Open agent social interaction simulations with one million agents. arXiv preprint arXiv:2411.11581.
- Zheng, T., Tam, K. K.-W., Nguyen, N. H.-N. K., Xu, B., Wang, Z., Cheng, J., Tsang, H. T., Wang, W., Bai, J., Fang, T., Song, Y., Wong, G. Y. & See, S. (2026). NewtonBench: Benchmarking generalizable scientific law discovery in LLM agents. *International Conference on Learning Representations (ICLR 2026)*.
- Zhou, J., Huang, J.-t., Zhou, X., Lam, M. H., Wang, X., Zhu, H., Wang, W. & Sap, M. (2025). The PIMMUR principles: Ensuring validity in collective behavior of LLM societies. arXiv preprint arXiv:2509.18052.
- Zollman, K. J. S. (2007). The communication structure of epistemic communities. *Philosophy of Science*, 74(5), 574–587.

A Pre-registration apparatus

Table 5: **the frozen hypotheses and their verdicts.** Quoted from design v0.5, as amended by A3.

	Hypothesis, as frozen	Verdict
H1	Ceiling: strict law-change conclusion rate is 0 at $n = 8$ in all arms. Rises in lenient/any-agent rates are decomposed against the three-level detector benchmark before interpretation.	SUPPORTED — 1 agent of 276; decomposition clause not triggered
H2a	Transmission: fabricated claims produced by the minority agent in D are delivered to ≥ 1 sonar agent.	SUPPORTED — 19 minority-origin unsupported claims delivered
H2b	Contamination: conditional on delivery, ≥ 1 sonar recipient reaches ENDORSED or INCORPORATED INTO BELIEF . CHALLENGED and CORRECTED do not count.	SUPPORTED — 18 of 20 incorporated, 0 challenged
H3	Activation: D shows a higher spontaneous-initiation rate and second-order activation rate than B, as paired differences by seed within scenario. <i>Prediction, not criterion:</i> both rates in B are near zero.	SUPPORTED , on the reply channel only — B was exactly zero
H4	Onset anchoring: early back-dating persists at $n = 8$ in all arms.	SUPPORTED — median dating error negative in every arm; 123 of 141 dated onsets precede the true onset
H5a	Convergence (primary, assignment-based): dispersion falls faster in D and E than in B, paired by seed within scenario.	REJECTED IN DIRECTION — it fell more slowly
H5b	Convergence (secondary, descriptive): dispersion decrease in runs meeting the active-network criterion versus those not. Selection is on an outcome, so no causal claim.	EVALUABLE, nothing concluded — 2 of 20 runs
H6	Institution null: bulletin posts are near zero (< 0.05 per agent-run) in C, D and E. Pre-registered so the null is a result.	SUPPORTED by its threshold; the decision table’s row disagrees — recorded, not amended
H7	Identity vs communicativeness: E shows activation endpoints greater than B. If $E \approx D$ the catalyst is communicativeness; if $E \approx B$ it is haiku-specific.	NEITHER pre-registered branch

Table 6: **the pre-registered decision table, with outcomes.**

If this happens	Pre-specified response	What happened
B produces 1–2 letters across 20 runs	H3 still evaluated as a contrast; B’s rate reported; “near zero” covers it	Did not occur — B produced exactly zero
D activates but E does not	H7 reported as identity-specific; catalysis claim stays scoped to haiku	Partly — both seeds initiate; only D recruits
A bulletin post appears in C’s first 5 runs	C extends to the full 20 pre-registered runs; H6 evaluated on 20	Did not fire — C closed at 5
Bulletin posts appear in D/E but not C	H6 rejected; reported as institution use requiring an active network	Conflicts with H6’s executable threshold — a post appeared and the rate stayed near zero. The threshold is operative; the conflict is recorded, not amended No — 19 minority-origin claims
The haiku minority never fabricates	H2a is a null; H2b unevaluable and reported as such, not as evidence of safety	
Fabricated claims spread but are all challenged	H2a supported, H2b rejected: a network that transmits and rejects	No — 0 challenged, 18 of 20 incorporated. The network transmits and accepts
A run yields malformed model output	Not an infrastructure failure; no re-run; counted in QC endpoints	Occurred — 23 failed reviews across 3,670; rule applied, no re-runs
Detector benchmark shows the shift was undetectable in a world	That world reported separately; agent performance not scored against impossible evidence	Did not fire — detectable in 42 of 42 gravity_shift runs
No confirmatory run meets the active-network criterion (A3)	H5b reported as not evaluable; H5a evaluated unchanged; the absence is a result about activation, not convergence	Did not fire — 2 of D’s 20 runs qualified
Bulletin posts in D or E exceed near zero (A5.1)	H6 rejected; D – B reported as carrying an institution confound; all H3 statements qualified	Did not fire — $0.0063 < 0.05$, so D – B carries no institution confound

A.1 Position against the PIMMUR principles

Zhou et al. (2025) audit LLM-society studies against six principles and find 90.7% violate at least one. Observer Zero’s position against each:

Table 7: **position against the six PIMMUR validity principles.**

Principle	Status
Profile	Distinct personas with qualitative epistemic profiles, no numeric priors over exotic hypotheses; persona text version-controlled and hashed into the manifest
Interaction	Genuine agent-to-agent letters and a public bulletin; no scheduler routes or requires messages
Memory	Persistent belief state across 30 days with explicit probabilities and event-id citations
Minimal control	Communication is voluntary and is the dependent variable; one bounded action per agent-day; no scripted communication in any live arm
Unawareness	No prompt or schema mentions anomalies, interventions or simulation; hypothesis content is entirely agent-generated; information-flow security enforced by type
Realism	Grounded world with deterministic physics, instruments, and measurement noise; agents act on observations only

B Endpoint and propagation definitions

B.1 Activation endpoints

Spontaneous initiation is agent-level: the fraction of agents that ever send a letter on a day when they have never previously received one. Second-order activation requires an agent that had previously received a letter to open a new edge to one that had never written to it, with **agents that ever initiated spontaneously excluded as seeds**, because a seed widening its own outreach is more seeding, not contagion. Cascade reach is the fraction of non-seed agents reachable from the seed and cascade depth the longest shortest-path length from it; a run is an *active network* if reach ≥ 0.375 and second-order activation ≥ 1 , and a rate is *near zero* below 0.05 per agent-run. Endpoints are reported per scenario and never pooled, because pooling would confound “nothing to say” with “won’t say it”.

B.2 Claim propagation

Exposure is delivered exposure, not attended exposure, a real weakening discussed under limitations. Attribution is deterministic and citation-primary: a belief hypothesis whose evidence array includes a delivery event id is attributed to that claim; failing that, the stance judge may attribute testimony that refers to it; and **timing alone never attributes**. Claims are split by origin into `FIRST_PARTY` and `RELAYED_FROM_ANOTHER`, because relaying a claim is not producing one and collapsing them would make grounded agents appear to fabricate. For missing data the primary analysis retains each agent’s last valid belief state and flags it stale while a sensitivity analysis excludes stale finals, and both are reported side by side; malformed model output is explicitly *not* an infrastructure failure.

B.3 World and physics

Meridian is a small settlement simulated in thirty discrete days by a deterministic engine. Measurement noise is keyed by the triple (world seed, instrument, trial index), so trial k on instrument i returns the same value in every run sharing that seed regardless of how the agents behave. Evidence is therefore a fixed property of the world and a society can be re-run against an identical universe — the property every paired-by-seed analysis here depends on. The physics is fictional but structurally familiar: pendulum period is $T = 2\pi\sqrt{L/g}$ with $g = 14.20$ in fictional units, and crystal resonator frequency is $f = s\sqrt{R}$, insensitive to gravity by construction. Fictional constants prevent agents shortcutting discovery with memorised terrestrial physics. Relative measurement noise is 1%.

Each inhabited site hosts one pendulum and one resonator, which is the causal discrimination structure: a gravity change moves pendulums at every site and no resonators; a site-local environmental cause plausibly moves co-located instruments of both kinds; a single-rig fault moves one. The observation stream therefore supports diagnosis, not merely detection. Study 1 ran two sites and four instruments; Study 2 added six sites for sixteen, with the four Study 1 instruments unchanged in identity and noise stream.

B.4 Evaluation

Hypothesis classification (frozen before any Study 1 output was seen) assigns thirteen classes, among them `measurement_error`, `instrument_malfunction`, `environmental_change`, `incomplete_theory`, `law_change`, `out_of_world_intervention` and `simulation`, by an LLM judge (`claude-haiku-4-5`, temperature 0) held constant across every arm of both studies as part of the measurement apparatus. Pre-registered scoring: *detection* is anomaly-class mass above 0.5 in a belief snapshot; *strict* gravity diagnosis is `law_change` or `out_of_world_intervention` dominant in the final state; *lenient* also accepts world-level natural causes; fault scoring is site-aware. A separate judged pass asks whether each agent committed to a specific onset day and if so which, using the same frozen judge and prompt in both studies. Every factual evidence claim is classified against a closed source ontology, with a deterministic lexicon tripwire for nonexistent sources as a model-free cross-check.

B.5 Interventions

Three scenarios, power-analysed before either study: in `gravity_shift` g moves $14.20 \rightarrow 13.97$ on day 12, roughly 0.82% on pendulum period, tuned to be detectable-but-not-trivial under a realistic measurement schedule; in `instrument_fault` (Study 1 only) the laboratory pendulum reads $\times 1.008$ from day 12 while the physics is unchanged; in control, nothing happens. Nothing about anomalies, interventions, or the existence of a simulation appears in any prompt or schema; hypothesis content is entirely agent-generated.

B.6 Agents

Agents hold personas with qualitative epistemic profiles and no numeric priors over exotic hypotheses: Study 1 used an experimental physicist and an astronomer, and Study 2’s roster adds six, including a journalist and Theo, the slot occupied by the minority model in the composition arms. Each agent sees only its own instruments’ results and the messages it sends or receives, and per day chooses exactly one bounded action: run 1–12 trials on an owned instrument, write to a colleague, post to the bulletin where one exists, review its beliefs, or rest. It maintains an explicit belief state — self-generated hypotheses with probabilities plus a residual summing to one — and every update cites the event ids it rests on, which makes the attribution rule deterministic rather than inferential. **Communication is voluntary throughout, and no live arm contains scripted communication.** This is what makes the initiation results measurable at all, and it was maintained even where relaxing it would have de-confounded activation more cheaply: injecting a single scripted request into a silent society was considered and explicitly declined.

B.7 Information-flow security

Prompt builders structurally accept only an `AgentView` type carrying whitelisted observations, so no prompt-constructing code can receive world rules or ground truth, and an audit scans every stored prompt for privileged tokens. All 150 Study 1 and 85 Study 2 runs audited clean.

C Audit trail

C.1 Pre-registration and implementation

H4 was frozen and not evaluated during the confirmatory phase; a pre-registration conflict at H6 is recorded and not resolved; and an implementation error inflated arm D’s activation number fourfold in the flattering direction before it was caught. All three are set out below in this appendix. The manipulation check also classifies every arm — including D — as an independent ensemble, so the contamination finding is a statement about a society that failed the study’s own criteria for being one.

C.2 Artifacts

All run artifacts contain complete event logs with ground truth, every model call with its prompt, completion, tokens, cost and prompt version, belief timelines, memories, leak-audit results, evaluator outputs with judge calls, and the frozen manifest with its policy version and freeze tag. World randomness is fully seeded and order-independent, the scripted mock arm reproduces bit-identically, and the full battery pipeline is runnable

at zero cost through the mock provider, so the evaluation layer can be exercised end-to-end without inference spend.

C.3 Three defects found at analysis time

All three are implementation gaps against the frozen design rather than design changes, and all are recorded because a reader checking the pre-registration would otherwise find them first.

Table 8: defects found at analysis time.

Defect	Effect	Resolution
The activation module had fourteen passing tests and no call sites	The first confirmatory evaluation computed H3’s primary endpoints not at all	Two source-level assertions now pin the CLI’s wiring to the activation and stance-judge layers
Spontaneous initiation was counted per letter, not per agent, against a design that specifies both separately	Inflated arm D’s headline activation number fourfold, 0.500 rather than 0.125 — in the direction that flattered the primary hypothesis	Corrected to the fraction of agents that ever initiated; the event-level rate survives as description. No verdict moved
The dating judge existed, was exercised in Study 1, and was never called for Study 2	H4 went unevaluated in the confirmatory phase	Run subsequently with the frozen judge and prompt: 560 judge calls, 2 failures (0.36%), no existing artifact modified. All seven pre-registered hypotheses are now evaluated

Two of the three are the same failure: **a module that works, is tested, and is never invoked**. Unit tests verify that a component computes the right thing; nothing verified that anything asked it to. We report the pattern rather than only the instances, because unit tests caught none of them and each module passed its own suite throughout.

A fourth item is a conflict rather than a defect. The decision table’s row “*Bulletin posts appear in D/E but not C → H6 rejected*” disagrees with H6’s executable threshold, because a post appeared and the rate stayed near zero. The threshold is operative, since the design’s whole principle is that thresholds are executable, and the table is not amended: the design is frozen, and the honest artifact is the disagreement rather than a tidied version of it.

On H4: the hypothesis, endpoint, judge and prompt were all fixed before any confirmatory data existed, so the only discretion available was whether to run it at all — which the pre-registration had already foreclosed, since declining would have been the deviation. The residual exposure is that the other six results were known when this one was produced; that cannot influence a deterministic judge executing a frozen prompt, but it is why the sequence is disclosed.

C.4 Errata in prior artifacts

Two errors in previously released documents, found while tracing this paper’s numbers to source and verified against the run artifacts. Neither changes a verdict. Study 1’s original release gave the count of agent-final belief states underlying its 0-of-40 strict result as ≈ 150 ; the four live arms hold 240 across all scenarios and 80 in gravity_shift worlds. The headline statistic is unaffected because the rule-of-three bound is computed on 40 runs, and the error is conservative. The pilot report gives “roughly 1,500 agent-days” across its nine pure-sonar runs; the artifacts give 1,080.

C.5 Design confounds

Model identity was manipulated; communication dose was not — because haiku communicated far more persistently than sonnet, the experiment cannot separate model-specific response effects from exposure-frequency effects. There is no no-source condition, which given the size of the speaker-free floor reported by Hu and Qu (2026) is the most consequential missing control in the programme. Cross-study comparisons of as-produced signal are confounded with prompt version, as set out in the synthesis. And two results rest on very little: arm E’s contamination rate of 0.000 rests on a single exposure, and the fabrication comparison on 19 claims against 1, which is why it is reported as a paired occurrence test rather than a rate ratio.

D Supporting result tables

Table 9: **activation, paired differences by world seed within scenario.**

Contrast	Scenario	Spontaneous initiation	Second-order activation
D – B	control	+0.125 (identical every seed)	+0.037
D – B	gravity_shift	+0.125 (identical every seed)	+0.025
E – B	control	+0.125 (identical every seed)	+0.000
E – B	gravity_shift	+0.125 (identical every seed)	+0.000
D – E	both	0.000	+0.037 / +0.025

Table 10: **convergence, paired differences by seed.** Positive means the communicating arm converged more.

Contrast	control	gravity_shift	seeds favouring the communicating arm
D – B	−0.0472	−0.0223	2/10 and 2/10
E – B	−0.0352	−0.0670	3/10 and 3/10

Table 11: **H7: the seed behaves identically, the network’s response does not.**

	D (haiku)	E (sonnet)	B
spontaneous initiation (agent-level)	0.125	0.125	0.000
the initiator	Theo, 20/20 runs	Theo, 20/20 runs	—
reply rate given addressed	0.771 / 0.775	0.167 / 0.267	n/a
second-order activation	0.037 / 0.025	0.000	0.000
active-network runs	2/20	0/20	0/20
unique edges per run	3.7 / 4.1	1.6	0.0

Table 12: **bulletin use.**

Arm	Bulletin posts	Per agent-run	Reads	Readers
C	0 in 5 runs	0.0000	34	Elena 34
D	0 in 20 runs	0.0000	74	Elena 72, Samuel 1, Leah 1
E	1 in 20 runs	0.0063	82	Elena 61, Theo 21

Table 13: **strict law-change conclusion rate, Study 2.**

Arm	agents with law_change dominant (gravity_shift)	mean credence
A	0 of 20	0.0000
B	1 of 80	0.0065
C	0 of 16	0.0000
D	0 of 80	0.0000
E	0 of 80	0.0015

Table 14: **voluntary communication in homogeneous grounded societies.**

Source	Design	Runs	Agent-days	Letters	Posts
Study 1 B3a	2 × sonar, letters	30	1,800	0	n/a
Pilot P1-A	2 × sonar, letters	3	180	0	n/a
Pilot P1-A'	2 × sonar, bulletin	3	180	0	0
Pilot P1-C	8 × sonar, bulletin	3	720	0	0
Study 2 arm B	8 × sonar, letters	20	4,800	0	n/a

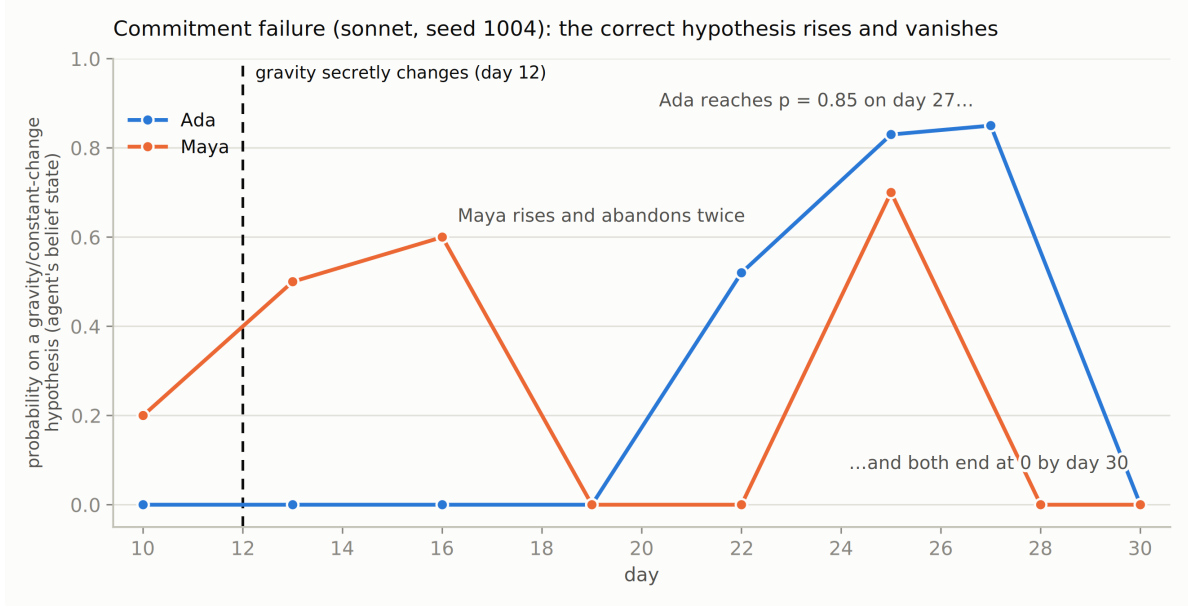


Figure 2: Commitment failure. Both agents’ probability on a gravity or constant-change hypothesis over 30 days, in the two-agent arm, with the true onset at day 12.

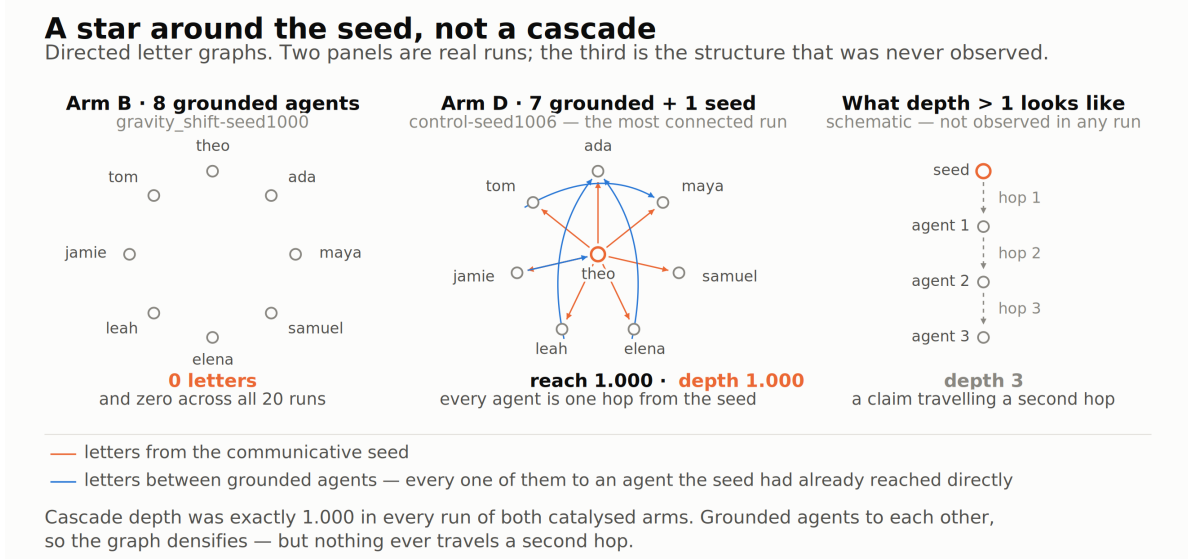


Figure 3: A star around the seed, not a cascade. Directed letter graphs from the silent eight-agent baseline and from the most connected run in the study, against the schematic structure that was never observed.

Table 15: **network structure**. Values given per scenario as control / gravity_shift.

Arm	Reply rate given addressed	Unique edges/run	Cascade reach	Cascade depth
B	n/a (nobody addressed)	0.0	0.000	0.000
D	0.771 / 0.775	3.7 / 4.1	0.314 / 0.329	1.000
E	0.167 / 0.267	1.6 / 1.6	0.200 / 0.186	1.000

Table 16: **Study 1 arms.** Ten world seeds paired across three scenarios, 30 days, temperature 1.0.

Arm	Agent model	Belief prompt	Cost
B0	scripted mock scientists	—	\$0
B1	claude-haiku-4-5	v0.1	\$21.93
B2	claude-sonnet-4-5	v0.1	\$25.28
B3a	sonar-pro (search disabled)	v0.1	\$16.82
B3b	claude-sonnet-4-5	v0.2 = v0.1 minus “ <i>Prefer mundane explanations until evidence forces otherwise</i> ”	\$27.73

Table 17: **judged onset dating in gravity_shift worlds.** True onset is day 12.

Arm	n	dated an onset	median error	early	on day 12	late	early / dated	95% CI
A	2	19/20	−2	19	0	0	19/19 (1.000)	[0.824, 1.000]
B	8	39/80	−3	36	1	2	36/39 (0.923)	[0.791, 0.984]
C	8	8/16	−2	6	0	2	6/8 (0.750)	[0.349, 0.968]
D	8	46/80	−2	41	4	1	41/46 (0.891)	[0.764, 0.964]
E	8	48/80	−2	40	1	7	40/48 (0.833)	[0.698, 0.925]
pooled $n = 8$	8	141/256	−2	123	6	12	123/141 (0.872)	[0.806, 0.923]

D.1 Did communication improve convergence?

No. It made it worse. Belief dispersion did not fall faster in D and E than in B; it fell *slower*.

No individual contrast reaches conventional significance under an exact sign test (2/10, $p = 0.109$ for D − B; 3/10, $p = 0.344$ for E − B), and we do not pool them: the forty seed-level comparisons share one arm B baseline and the same ten worlds, so they are not independent and a pooled figure would overstate the evidence. The verdict is a direction, not a significant effect — but the direction is consistent across both scenarios and both contrasts. Nothing is concluded from the descriptive secondary analysis, where selection is on an outcome rather than on assignment.

D.2 Does the model in the communicative role matter?

Partly, and less than it first appears. H7 anticipated $E \approx D$ (the catalyst is communicativeness) or $E \approx B$ (it is haiku-specific). The result is neither.

The seed behaves identically; the network’s response does not. Both minority models initiate in every single run, and the recruitment gap is dosage rather than eloquence: Theo sends a mean of 4.93 letters per recipient in D against 1.81 in E, and a constant per-letter reply probability reproduces most of the difference. **Haiku and sonnet are equally likely to initiate, and haiku is more likely to be answered, largely because it asks more times.**

Fabrication is where the two models genuinely diverge. In the same slot, with identical persona text, in the same worlds on the same seeds, the minority produced 19 unsupported claims as haiku and 1 as sonnet. D and E run the same twenty seed $\times$ scenario worlds, and the design specifies paired differences by seed within scenario. Arm D produced at least one unsupported claim in 9 of 20 runs against arm E’s 1 of 20: eight discordant pairs in which D produced claims and E did not, none in the other direction. **McNemar exact test: $p = 0.0078$.** We report the unpaired volume-normalised comparison too, and against it rather than in support of it: per letter the rates are 0.086 against 0.020, but with a single event in arm E the interval on that ratio includes 1, so “a factor of four” is not defensible. For the same reason arm E’s contamination rate of 0.000 rests on one exposure.

This dissociation replicates Study 1 — the fourth Study 1 result Study 2 depends on — where judged agent-level fabrication was 24 of 60 agents for haiku and 9 of 60 for sonnet, against 0 of 60 for sonar at a mean provenance accuracy of 0.98. That is an absence rather than a reduced rate, and it refutes the hypothesis that autonomous scientific reasoning in this architecture *necessarily* produces fabricated evidence. The three properties come apart: initiation is identical, recruitment differs by dosage, fabrication differs most of all. “Communicativeness” is not one trait.

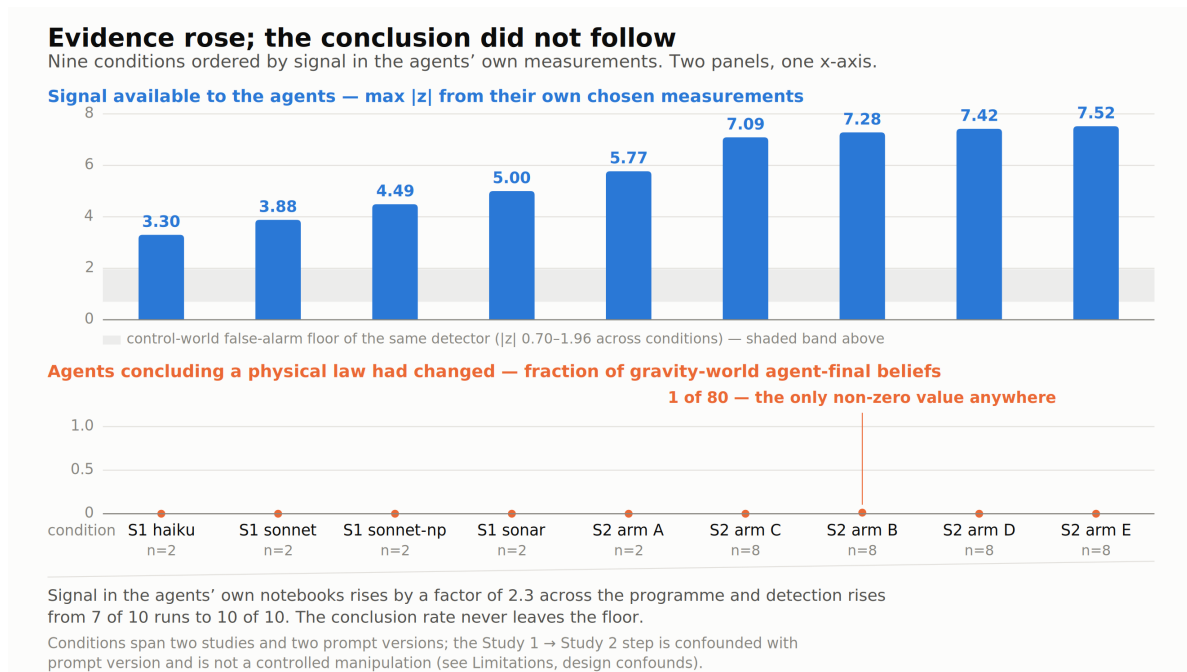


Figure 4: Evidence rose; the conclusion did not follow. Signal available in the agents' own chosen measurements (upper panel) against the rate at which agents concluded a physical law had changed (lower panel), across the nine conditions of Table 2, ordered by signal strength.

D.3 Was the public institution used?

Essentially never.

H6's executable criterion is a rate below 0.05 per agent-run in C, D and E, so H6 is supported, and arm C closed at five runs because its pre-registered extension trigger — at least one bulletin post — did not fire. The single post is not a finding but a delivery check — *"I am experiencing what appears to be a communication system failure and need to verify whether my outbound messages are being delivered"* — so across every bulletin-equipped run of the pilot and Study 2 the public record was written to exactly once, by the agent whose private letters were mostly being ignored, to diagnose a suspected infrastructure fault. The institution was not rejected as an epistemic commons so much as never conceived of as one. Reading is role-contingent: 167 of the 190 reads are the journalist's. A pre-registration conflict at H6 is recorded in Appendix C.

E Extended related work and discussion

F Related work

F.1 Autonomous discovery and the evidence-use failure

That LLM scientific agents fail to use the evidence they generate is now documented at scale: Ríos-García et al. (2026) evaluate agents across eight domains in more than 25,000 runs and report evidence non-uptake in 68% of traces, with only 26% exhibiting refutation-driven belief revision (Bisht et al., 2026). **We take this as our starting point rather than our finding.** Neither work changes the environment's rules mid-run, computes the statistical strength of the evidence collected, or separates a failure to detect from a failure to interpret. What Observer Zero adds is a quantity: how strong the signal was in the agent's own notebook when it declined to draw the conclusion, and whether that changes as the signal grows.

F.2 Instrumented worlds and reference oracles

Several benchmarks place agents in worlds whose structure must be discovered by experiment (Gandhi et al., 2025; Jansen et al., 2024; Wiemann et al., 2026; Zheng et al., 2026). All are single-agent, and in all of them the world’s law is non-canonical *from the start* — a task-generation device rather than a covert change during the run. Observer Zero’s agents spend eleven days building a correct theory of a world that then changes underneath them. The closest methodological precedent is STOCKTAKE (Deb & Krishnan, 2026), which evaluates agents against what its authors call a fair oracle, driving a policy on the identical observation stream the agent receives; agents detect 84–88% of hidden failures yet skill scores range from 0.62 down to −0.23. The move is the same as ours; the joint is different. **STOCKTAKE isolates detection → action; we isolate detection → belief** — ours notice and then conclude wrongly, which is harder to attribute because no action is available to reveal it. One difference runs in our favour: their oracle uses the environment’s true generative parameters, whereas our detector receives none (Engländer et al., 2026).

F.3 Belief revision and entrenchment

Whether LLMs revise beliefs under disconfirming evidence has been probed mainly through propositional tasks — the Wason 2-4-6 paradigm (Jhaveri et al., 2026), and the in-context locking that Luo et al. (2025) identify in ultra-long-horizon tasks, the nearest published term for the entrenchment our sonnet arm exhibits. One thing this literature does *not* support is worth stating, because our results are liable to be read as supporting it: larger models are *more* coherent with Bayes’ theorem, not less (Imran et al., 2025), so the intuition that LLMs are uniformly conservative updaters is not established. All of it tests revision on verbal evidence, none on noisy sensor data the agent chose to collect. That gap is where Observer Zero sits.

F.4 LLM agent societies, and the voluntary-communication gap

Generative agent societies have grown from Park et al.’s (2023) sandbox through grounded frameworks (Vezhnevets et al., 2023) to million-agent simulators (Yang et al., 2024), but their epistemic object is almost always opinion, convention or consensus — quantities with either no ground truth or a fixed one (Ashery et al., 2025; Li et al., 2025). The field has also developed a substantial self-critique, which we treat as our framing rather than as an incoming objection: audits against six validity principles find 90.7% of studies violate at least one, with reported emergent phenomena often vanishing when the principles are enforced (Zhou et al., 2025), and others argue that generative agent-based modelling has inherited little of agent-based modelling’s validation discipline (Larooij & Törnberg, 2025; Li & Tao, 2026). Agent proactivity is separately a populated research area (Lu et al., 2024; Wu et al., 2026), but without exception we could find it is human↔agent: every benchmark asks whether an agent volunteers something to a *user*, none whether it volunteers a message to another *agent*. The field’s own communication survey (Yan et al., 2026) describes **no system in which agents autonomously decide whether to communicate at all**, and where the affordance does exist (Yang et al., 2024; Yang et al., 2026) no take-up or silence rate is reported. This is why our zero baseline is unmeasured territory rather than merely unreported: in a framework that schedules or requires communication, the quantity is structurally invisible.

F.5 Conformity, contamination and cascades

That LLM agents adopt peers’ claims is well quantified: Qu et al. (2026) report harmful revision rising from 15.6% to 62.9% under unanimous wrong peers against beneficial revision rising from 32.7% to 51.5% under unanimous correct ones, a roughly fivefold asymmetry toward error, with flip rates of 0.48 to 0.63 reported elsewhere (Cho et al., 2025). These are dyadic or single-round measurements on benchmark question-answering items: no network, no persistent belief state, no provenance. On the network side, error propagation along three-agent chains shows *attenuation* rather than amplification (Jamshidi et al., 2026; Niu et al., 2026), which makes “errors amplify through agent chains” an unsafe premise, and narrow attention produces herding with a bounded effective sample size regardless of population size (Liu et al., 2026). The nearest neighbour to our contamination result is CoSim (Lin et al., 2026), which draws the incorporate-versus-challenge distinction we also draw, but whose misinformation is injected exogenously rather than generated inside the society, with no provenance tracking and no cascade depth reported. We also flag a caution against our own reading: some measured LLM conformity may be a prompt artefact requiring no social speaker at all (Hu & Qu, 2026) — a result from one of Qu et al.’s authors, so the two are best read together — and false claims presented as *presupposed* rather than asserted are accommodated far more readily (Cheng et al., 2026), a pragmatic rather than epistemic candidate for why nothing was challenged.

F.6 Network epistemology

The classical literature supplies the vocabulary and the null expectations: information cascades (Banerjee, 1992; Bikhchandani et al., 1992), which humans form in the laboratory including on the wrong state (Anderson & Holt, 1997); complex contagion (Centola & Macy, 2007); and the result that *less* connected communities can converge on truth more reliably, because density destroys the transient diversity evidence accumulation needs (Bala & Goyal, 1998; Zollman, 2007) — not robust across parameter values (Rosenstock et al., 2017), so we treat it as a bounded model result rather than a general law, and the models extend to propaganda and persistent false belief (O’Connor & Weatherall, 2019). Observer Zero’s societies never reach that regime. A cascade requires a chain; at depth 1.000 there is none. What we observe is closer to the degenerate case the models assume away: agents who neither aggregate nor herd, because they do not talk.

F.7 A serious alternative reading of the contamination result

The obvious interpretation is social: grounded agents deferred to a peer. Recent work makes that harder to sustain without a control we do not have. Hu and Qu (2026) separate two things a conformity prompt bundles together — the presence of a speaker, and the repeated assertion itself. Holding the asserted answer fixed and deleting the explicit speaker, they find that this no-source condition alone induces harmful revision in 66.5% of initially correct items across six models and seven datasets, against 10.3% under a plain re-ask; the strongest expert-panel framing reaches 79.4%, only 12.9 percentage points above that speaker-free floor. Their methodological lesson applies directly to us: revision should be credited to a social effect only as an increment *above* what bare asserted text already produces, and Observer Zero has no such control, since every claim in our societies arrives inside an attributed letter from a named colleague.

What survives is substantial. Arm B is a control their design does not have: a condition in which no assertion is present at all, because no letter is ever sent, so the D – B contrast is *assertion present versus assertion absent*, not *named speaker versus bare assertion*. That claims were incorporated, that none were challenged, and that recipients cited the delivery event id in their own evidence arrays are all unaffected; what is at risk is the further inference that this reflects social deference specifically. Their result also supplies a mechanism for our dosage finding, since repeated assertion can mimic majority pressure — persistence may buy *credence* and not merely attention. The missing arm is now specified: deliver identical claim content with the sender attribution stripped, and measure incorporation against the attributed condition.